



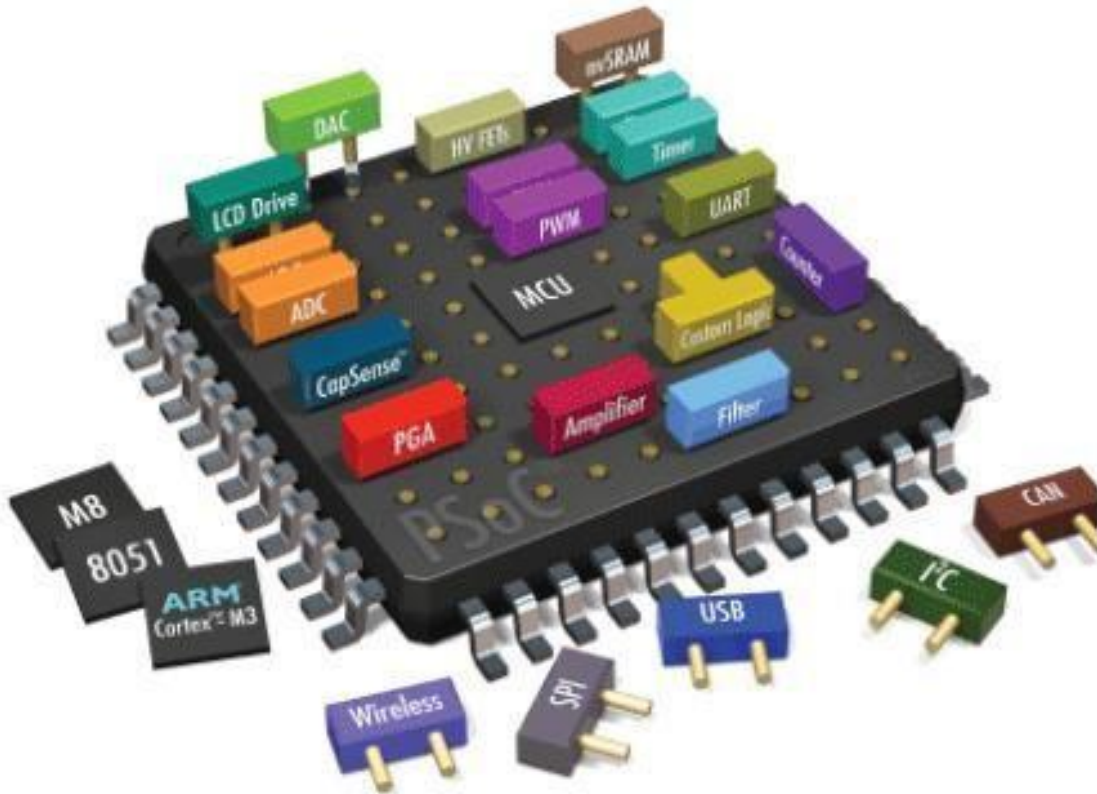
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Lab # 10: Understanding and practice of Step Motor Control and Photo Interrupter Control.

Embedded Systems

Understanding and practice of Step Motor Control and Photo Interrupter Control.



Dated:

Week 10

Semester:

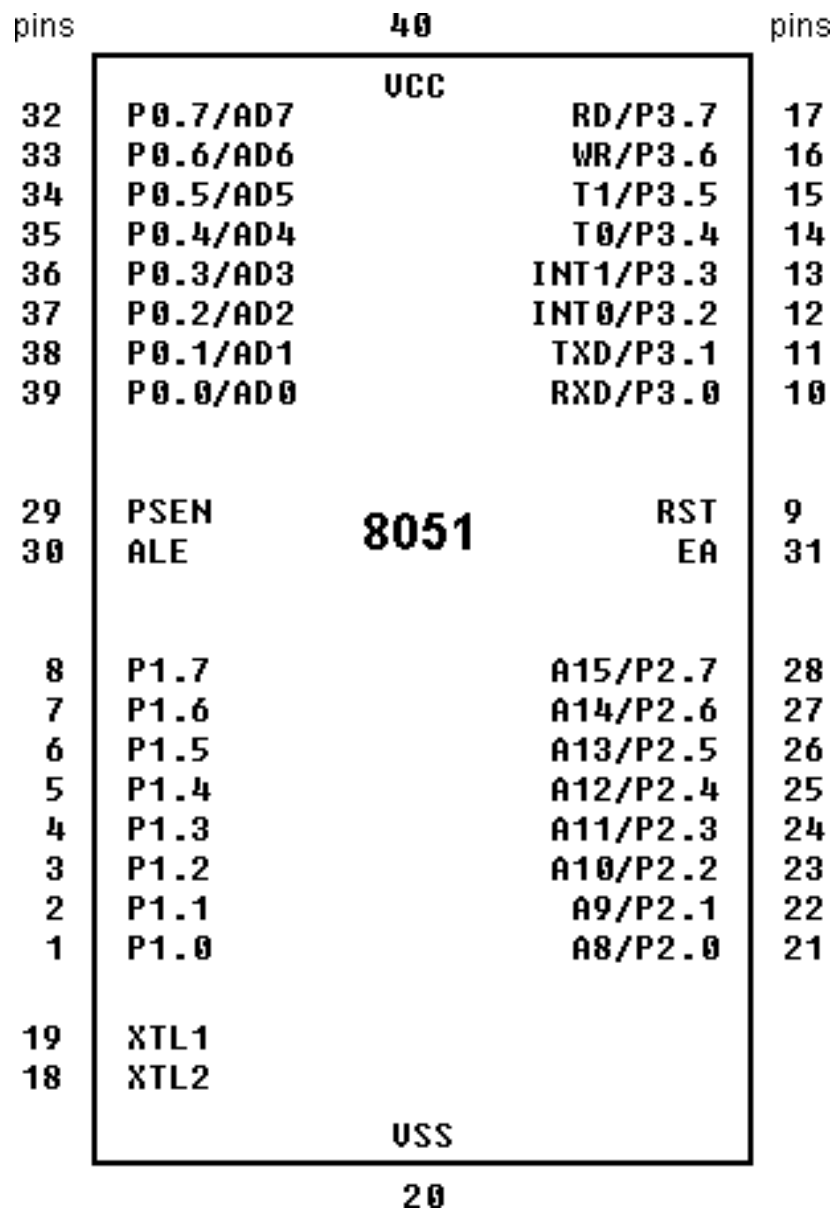
Fall 2022



**Lab # 10: Understanding and practice of Step Motor Control and
Photo Interrupter Control.**

8051 microcontroller:

8051 microcontroller is designed by Intel in 1981. It is an 8-bit microcontroller. It is built with 40 pins DIP (dual inline package), 4kb of ROM storage and 128 bytes of RAM storage, 2 16-bit timers. It consists of are four parallel 8-bit ports, which are programmable as well as addressable as per the requirement. An on-chip crystal oscillator is integrated in the microcontroller having crystal frequency of 12 MHz.





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- **Pins 1 to 8** – These pins are known as Port 1. This port doesn't serve any other functions. It is internally pulled up, bi-directional I/O port.
- **Pin 9** – It is a RESET pin, which is used to reset the microcontroller to its initial values.
- **Pins 10 to 17** – These pins are known as Port 3. This port serves some functions like interrupts, timer input, control signals, serial communication signals RxD and TxD, etc.

Pin	Name	Bit Address	Function
P3.0	RXD	B0H	Receive data for serial port
P3.1	TXD	B1H	Transmit data for serial port
P3.2	INT0-bar	B2H	External interrupt 0
P3.3	INT1-bar	B3H	External interrupt 1
P3.4	T0	B4H	Timer/counter 0 external input
P3.5	T1	B5H	Timer/counter 1 external input
P3.6	WR-bar	B6H	External data memory write strobe
P3.7	RD-bar	B7H	External data memory read strobe

- **Pins 18 & 19** – These pins are used for interfacing an external crystal to get the system clock.
- **Pin 20** – This pin provides the power supply to the circuit.
- **Pins 21 to 28** – These pins are known as Port 2. It serves as I/O port. Higher order address bus signals are also multiplexed using this port.
- **Pin 29** – This is PSEN pin which stands for Program Store Enable. It is used to read a signal from the external program memory.
- **Pin 30** – This is EA pin which stands for External Access input. It is used to enable/disable the external memory interfacing.
- **Pin 31** – This is ALE pin which stands for Address Latch Enable. It is used to demultiplex the address-data signal of port.
- **Pins 32 to 39** – These pins are known as Port 0. It serves as I/O port. Lower order address and data bus signals are multiplexed using this port.
- **Pin 40** – This pin is used to provide power supply to the circuit.



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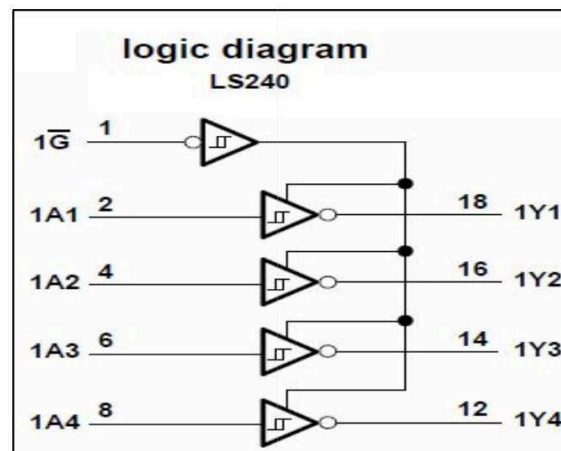
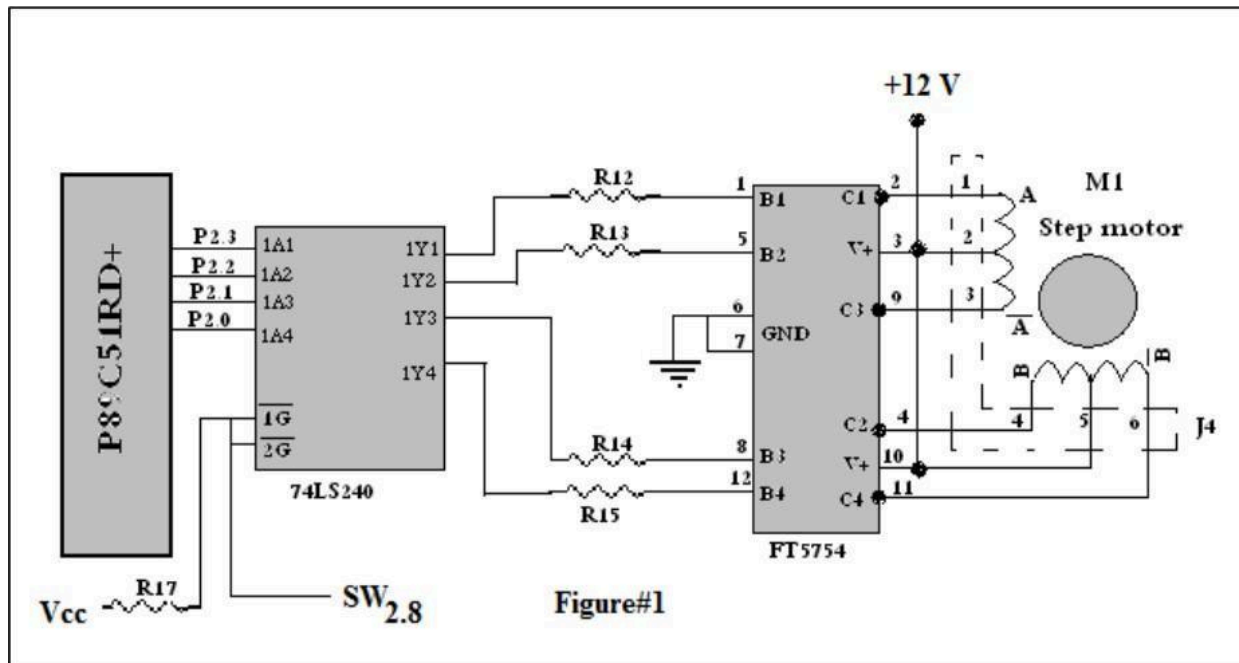
Step Motor Control.

Objective:

1. This experiment demonstrate the control of the 4-phase step motor M1 on MTS-51 trainer.
2. Describe the operation and switching schemes of the step motor.

Circuit Description:

74LS240 IC is used to invert the input signal and to promote the driving capability of 8051 Port 2. The enable inputs 1G and 2G are controlled by the switch SW2-8.





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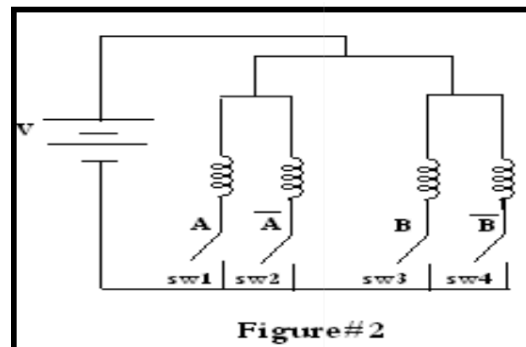
When this switch is in off position, $1G = 2G = 1$, the outputs of 74LS240 are in high impedance state so that the motor stops (according to the logic diagram of 74LS240).

The port 2 lines (P2.0-P2.3) are connected to the A, B, A and B coils of step motor, respectively. Since 74LS240 is an inverter, the corresponding coil will be energized if the switching signal is 0.

The FT5457 contains four Darlington pairs act as electronic switches to switch motor coils on or off.

Step Motor Operation

- Step Motors are electromagnetic incremental motion actuators, which convert digital pulse inputs to analog output motion. One pulse advances the rotor shaft one step increment (step angle) and latches it magnetically at the precise point to which it is stepped.
- Figure #2 shows the connection of the step motor for uni-polar drive. These power transistors receive the digital pulses to input terminals and energize motor coils.



Figure#2

- Switching Schemes: There are three switching schemes: 1-Phase-on switching, 2-Phase-on switching, and 1-2-Phase-on switching.

□ 1- Phase-on Switching: Table#1 shows the truth table of 1-Phase-on switching scheme. In this scheme, one pulse advances the rotor one step angle.

Step	Coil Energized				Direction of Rotation	
	A	B	A	B	CW	CCW
1	1	0	0	0		
2	0	1	0	0		
3	0	0	1	0		
4	0	0	0	1		
5	1	0	0	0		

Table#1



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□ **2- Phase-on Switching:** Table#2 shows the truth table of 2-Phase-on switching. In this scheme, one pulse advances the rotor one step angle.

Step	Coil Energized				Direction of Rotation	
	A	B	A	B	CW	CCW
1	1	1	0	0		
2	0	1	1	0		
3	0	0	1	1		
4	1	0	0	1		
5	1	1	0	0		

Table#2

□ **1-2-Phase-on switching**

The 1-2-Phase-on switching shown in Table#3 below is also called half-stepping switching scheme. One pulse advances the rotor half the step angle.

Step	Coil Energized				Direction of Rotation	
	A	B	A	B	CW	CCW
1	1	0	0	0		
2	1	1	0	0		
3	0	1	0	0		
4	0	1	1	0		
5	0	0	1	0		
6	0	0	1	1		
7	0	0	0	1		
8	1	0	0	1		
9	1	0	0	0		
10	1	1	0	0		

Table#3

Exercise#1

Functional Description:

Drive the step motor to rotate CW using two phase-on switching.

Source Program:



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```
File Edit Format View Help
      ORG      000H

START:
      MOV      P2,#00110011B
      CALL     DELAY

      MOV      P2,#10011001B
      CALL     DELAY

      MOV      P2,#11001100B
      CALL     DELAY

      MOV      P2,#01100110B
      CALL     DELAY

      JMP      START
=====
; DELAY 20mS
=====
DELAY:
      MOV      R6,#40
DL1:   MOV      R7,#249
      DJNZ     R7,$
      DJNZ     R6,DL1
      RET
      END
```




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Photo-Interrupter Control

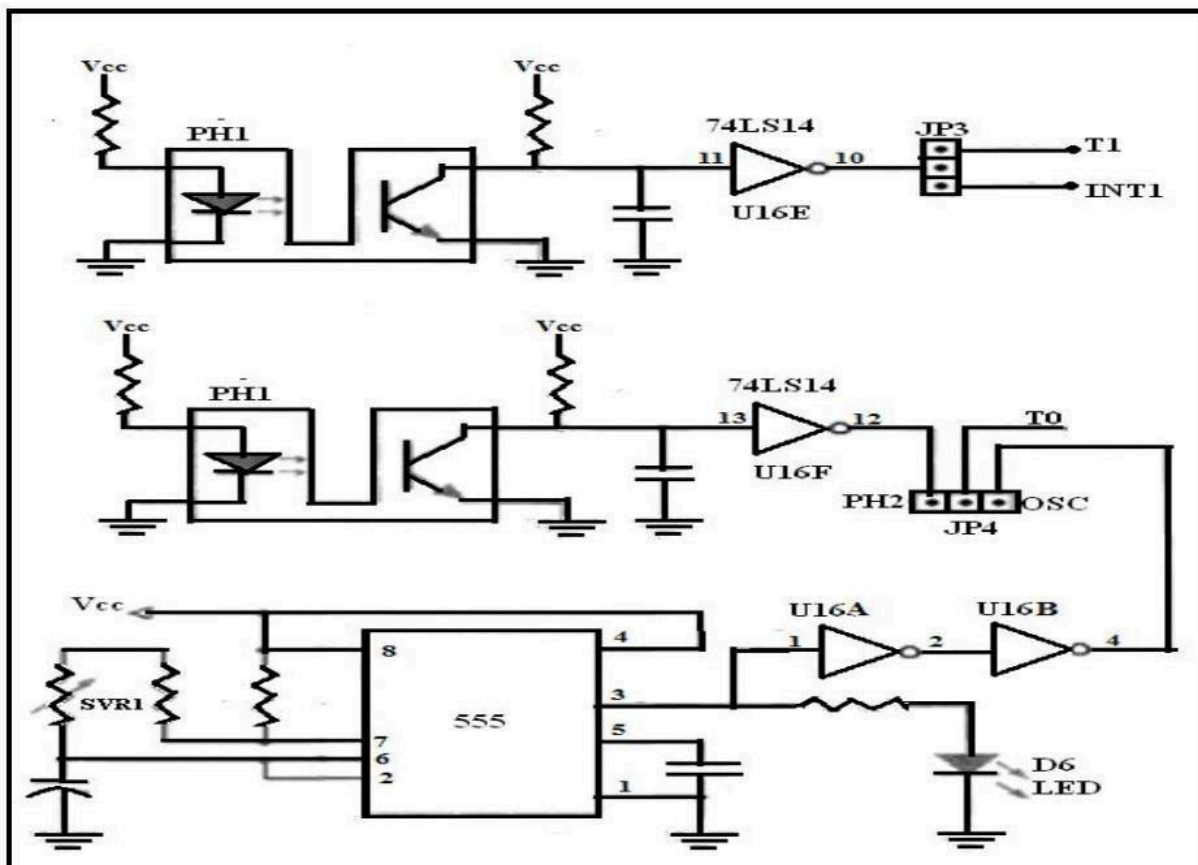
Objective:

1. Study the schematic diagram of photo interrupter (PH1, PH2) and the pulse generator circuit.
2. To demonstrate the control applications of photo-interrupter, for example controlling the direction of rotation of the step motor.

Circuit Description:

The MTS-51 has two photo-interrupters PH1 and PH2. When light is blocked, the photo-transistor is cut-off and a low presents on the inverter output terminal.

When light is not blocked, the photo-transistor conducts in saturation and a high presents on the inverter output terminal.



PH1 output can be connected to T1 or INT1 using a jumper cap on JP3. PH2 output can be connected to T0 using a jumper cap on JP4.



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The output signal of pulse generator is connected to pin 3 (OSC) of JP4. Using a jumper cap, the signal to T0 input of 8051 can be selected from PH2 output or pulse oscillator output.

Exercise#2

Functional Description:

Use PH2 output to control the direction of rotation of the step motor. When PH2 is not blocked by an object (for example a piece of paper), the step motor rotates CW. If PH2 is blocked by an object, the step motor rotates CCW.

Source Program:

```
ORG      000H

TEST:    MOV      A,#00110011B
         JB       P3.4,TURNR
         RL       A
         MOV      P2,A
         CALL     DELAY
         JMP      TEST

TURNR:   RR       A
         MOV      P2,A
         CALL     DELAY
         JMP      TEST

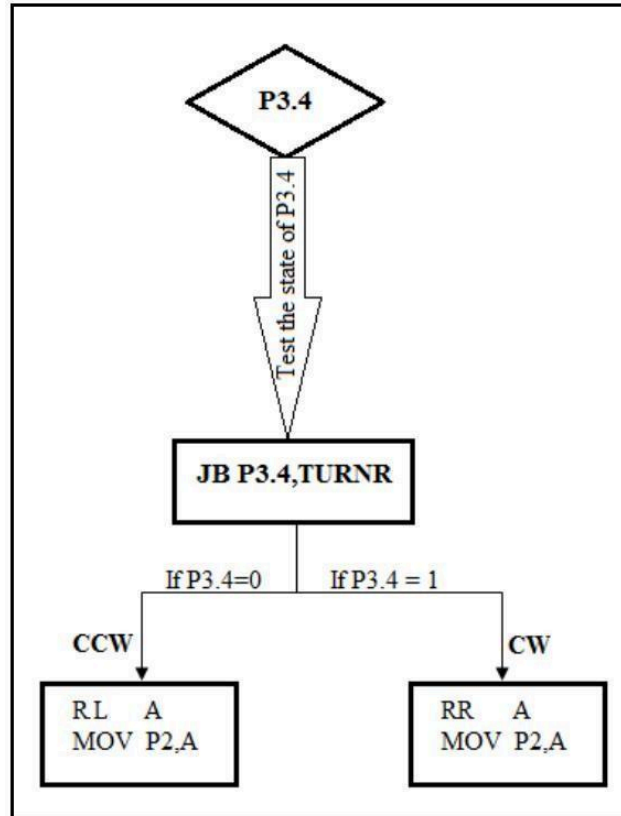
=====
; DELAY 10mS
=====
DELAY:
DL1:     MOV      R6,#20
         MOV      R7,#249
         DJNZ     R7,$
         DJNZ     R6,DL1
         RET

END
```



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Program Description



Lab Tasks:

1. Perform the example in exercise#1 and analyze the output and complete the following Table:

Loop #	Motor Coils				8051 Port 2			
	A	B	A	B	P _{2.0}	P _{2.1}	P _{2.2}	P _{2.3}
1	1	1	0	0				
2	0	1	1	0				
3								
4								

2. Write a program an assembly language code of the program in exercise#1 using RR instruction.



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3. Perform the example in exercise#1 and analyze the output and complete the following Table:

P3.4	Loop#	P _{2.0}	P _{2.1}	P _{2.2}	P _{2.3}
0	1	0	0	1	1
	2				
	3				
	4				
1	1	0	0	1	1
	2				
	3				
	4				

Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description about how its work (3-5 lines)
2. The code
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.